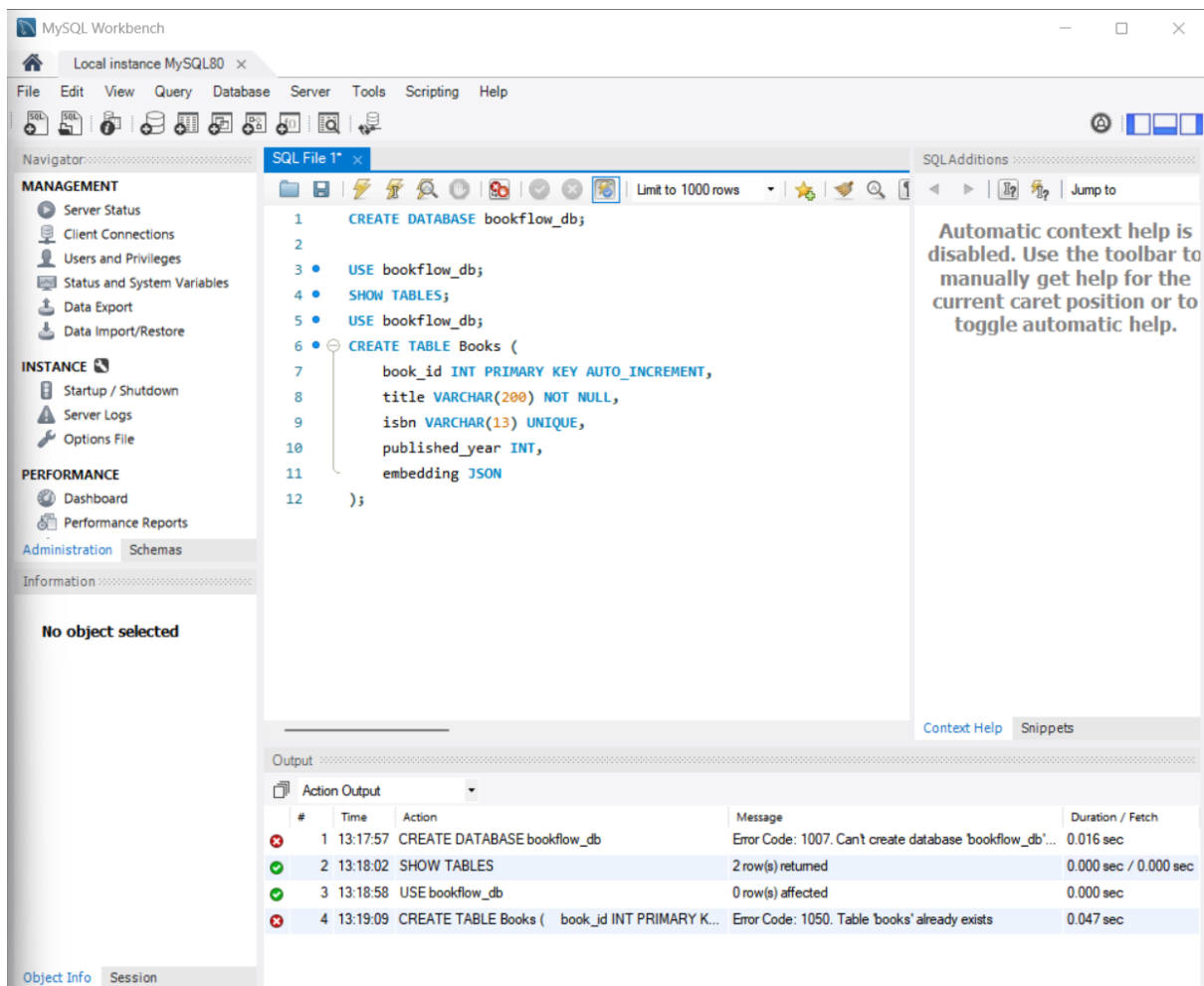


Semantic Search & API Design

Scenario: Users complain the library's search is too literal — searching "stories about space" should still surface relevant books even without an exact keyword match. The frontend team also needs an API contract to start building the search UI early.

Tasks: 1) Semantic Search Setup (Vector) — enable pgvector in bookflow_db and add an embedding column of type VECTOR(3) to the Books table. 2) Similarity Querying — insert mock vectors representing book themes and use the <-> (L2 distance) or <=> (cosine similarity) operator to find the 3 closest books to a given search vector. 3) OpenAPI Specification — design a REST spec for GET /v1/books/search with query parameters q and limit, a 200 OK response schema, and a 400 Bad Request error envelope. 4) Verifying REST Constraints — ensure the design is stateless and uses correct HTTP verbs.



The screenshot shows the MySQL Workbench interface. The left sidebar contains the Navigator pane with sections for MANAGEMENT, INSTANCE, PERFORMANCE, and Information. The main editor displays a SQL script with the following queries:

```
1 CREATE DATABASE bookflow_db;
2
3 USE bookflow_db;
4 SHOW TABLES;
5 USE bookflow_db;
6 CREATE TABLE Books (
7     book_id INT PRIMARY KEY AUTO_INCREMENT,
8     title VARCHAR(200) NOT NULL,
9     isbn VARCHAR(13) UNIQUE,
10    published_year INT,
11    embedding JSON
12 );
```

The right sidebar shows the SQLAdditions pane with a message: "Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help."

The bottom pane shows the Output window with the following table:

#	Time	Action	Message	Duration / Fetch
1	13:17:57	CREATE DATABASE bookflow_db	Error Code: 1007. Can't create database 'bookflow_db'...	0.016 sec
2	13:18:02	SHOW TABLES	2 row(s) returned	0.000 sec / 0.000 sec
3	13:18:58	USE bookflow_db	0 row(s) affected	0.000 sec
4	13:19:09	CREATE TABLE Books (book_id INT PRIMARY K...	Error Code: 1050. Table 'books' already exists	0.047 sec

File Edit View Query Database Server Tools Scripting Help

Navigator: SQL File 1* x SQLAddi

MANAGEMENT

- Server Status
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- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports

Administration Schemas

Information

No object selected

```
10 published_year INT,  
11 embedding JSON  
12 );  
13 USE bookflow_db;  
14  
15 DESCRIBE Books;  
16
```

Result Grid Filter Rows: Export: Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
book_id	int	NO	PRI	NULL	
title	varchar(100)	NO		NULL	
isbn	varchar(20)	YES	UNI	NULL	
published_year	int	YES		NULL	

Result Grid Form Editor Field Types

Auto disabled man current to

Navigator: SQL File 1* x SQLAdditions

MANAGEMENT

- Server Status
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- Status and System Variables
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- Data Import/Restore

INSTANCE

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PERFORMANCE

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Administration Schemas

Information

No object selected

```
10 published_year INT,  
11 embedding JSON  
12 );  
13 USE bookflow_db;  
14  
15 DESCRIBE Books;  
16 SELECT * FROM Books;
```

Result Grid Filter Rows: Edit: Export/Import

book_id	title	isbn	published_year
1	The Great Gatsby	9780743273565	1925
2	To Kill a Mockingbird	9780061120084	1960
3	1984	9780451524935	1949
4	Animal Farm	9780451526342	1945
*	NULL	NULL	NULL

Result Grid Form Editor Field Types

Automatic con disabled. Use t manually get current caret p toggle autor

FileEditViewQueryDatabaseServerToolsScriptingHelp

SQL File 1*

Limit to 1000 rows

Jump to

MANAGEMENT

Server Status

Client Connections

Users and Privileges

Status and System Variables

Data Export

Data Import/Restore

INSTANCE

Startup / Shutdown

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PERFORMANCE

Dashboard

Performance Reports

AdministrationSchemas

Information

No object selected

15 • DESCRIBE Books;

16 • SELECT * FROM Books;

17 • USE bookflow_db;

18

19 • ALTER TABLE Books

20 ADD COLUMN embedding JSON;

21 • DESCRIBE Books;

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

Result Grid

Form Editor

Field Types

Result 4 x

Read Only

Context Help

Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 3	13:18:58	USE bookflow_db	0 row(s) affected	0.000 sec
✗ 4	13:19:09	CREATE TABLE Books (book_id INT PRIMARY ...	Error Code: 1050. Table 'books' already exists	0.047 sec
✓ 5	13:20:01	DESCRIBE Books	4 row(s) returned	0.000 sec / 0.000 sec
✓ 6	13:20:36	SELECT * FROM Books LIMIT 0, 1000	4 row(s) returned	0.015 sec / 0.000 sec
✓ 7	13:21:10	ALTER TABLE Books ADD COLUMN embedding JS...	0 row(s) affected Records: 0 Duplicates: 0 Warning...	0.094 sec
✓ 8	13:21:20	DESCRIBE Books	5 row(s) returned	0.000 sec / 0.000 sec

Object Info

Session

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

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PERFORMANCE

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Administration Schemas

Information

No object selected

SQL File 1* x

```
16 • SELECT * FROM Books;
17 • USE bookflow_db;
18
19 • ALTER TABLE Books
20   ADD COLUMN embedding JSON;
21 • DESCRIBE Books;
22 • SELECT * FROM Books;
```

Limit to 1000 rows

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Result Grid

	book_id	title	isbn	published_year	embedding
▶	1	The Great Gatsby	9780743273565	1925	NULL
	2	To Kill a Mockingbird	9780061120084	1960	NULL
	3	1984	9780451524935	1949	NULL
	4	Animal Farm	9780451526342	1945	NULL
*	NULL	NULL	NULL	NULL	NULL

Books 5 x

Apply Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
4	13:19:09	CREATE TABLE Books (book_id INT PRIMARY ...	Error Code: 1050. Table 'books' already exists	0.047 sec
5	13:20:01	DESCRIBE Books	4 row(s) returned	0.000 sec / 0.000 sec
6	13:20:36	SELECT * FROM Books LIMIT 0, 1000	4 row(s) returned	0.015 sec / 0.000 sec
7	13:21:10	ALTER TABLE Books ADD COLUMN embedding JS...	0 row(s) affected Records: 0 Duplicates: 0 Warning...	0.094 sec
8	13:21:20	DESCRIBE Books	5 row(s) returned	0.000 sec / 0.000 sec
9	13:21:44	SELECT * FROM Books LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

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Administration Schemas

Information

No object selected

SQL File 1*

```
29 WHEN 3 THEN '[0.10, 0.80, 0.30]'
```

Execute the statement under the keyboard cursor

```
31 WHEN 5 THEN '[0.88, 0.12, 0.18]'
```

```
32 END
```

```
33 WHERE book_id IN (1, 2, 3, 4, 5);
```

```
34 SELECT book_id, title, embedding
```

```
35 FROM Books;
```

Result Grid

	book_id	title	embedding
1	1	The Great Gatsby	[0.9, 0.1, 0.2]
2	2	To Kill a Mockingbird	[0.85, 0.15, 0.25]
3	3	1984	[0.1, 0.8, 0.3]
4	4	Animal Farm	[0.2, 0.3, 0.9]
*	NULL	NULL	NULL

Books 7 x

Apply Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
7	13:21:10	ALTER TABLE Books ADD COLUMN embedding JS...	0 row(s) affected Records: 0 Duplicates: 0 Warning...	0.094 sec
8	13:21:20	DESCRIBE Books	5 row(s) returned	0.000 sec / 0.000 sec
9	13:21:44	SELECT * FROM Books LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec
10	13:22:13	SELECT book_id, title, embedding FROM Books LIM...	4 row(s) returned	0.000 sec / 0.000 sec
11	13:22:24	UPDATE Books SET embedding = CASE book_id ...	4 row(s) affected Rows matched: 4 Changed: 4 Wa...	0.031 sec
12	13:22:35	SELECT book_id, title, embedding FROM Books LIM...	4 row(s) returned	0.000 sec / 0.000 sec

MySQL Workbench

Local instance MySQL80

FileEditViewQueryDatabaseServerToolsScriptingHelp

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AdministrationSchemas

Information

No object selected

SQL File 1*

Limit to 1000 rows

48LIMIT 3;

49SELECT

50book_id,

51title,

52embedding

53FROM Books

54WHERE embedding IS NOT NULL;

Result Grid

Filter Rows:

Edit:

Export/Import

Result Grid

Form Editor

Field Types

book_id	title	embedding
1	The Great Gatsby	[0.9, 0.1, 0.2]
2	To Kill a Mockingbird	[0.85, 0.15, 0.25]
3	1984	[0.1, 0.8, 0.3]
4	Animal Farm	[0.2, 0.3, 0.9]
NULL	NULL	NULL

Books 9

Apply

Context Help

Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
9	13:21:44	SELECT * FROM Books LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec
10	13:22:13	SELECT book_id, title, embedding FROM Books LIM...	4 row(s) returned	0.000 sec / 0.000 sec
11	13:22:24	UPDATE Books SET embedding = CASE book_id ...	4 row(s) affected Rows matched: 4 Changed: 4 Wa...	0.031 sec
12	13:22:35	SELECT book_id, title, embedding FROM Books LIM...	4 row(s) returned	0.000 sec / 0.000 sec
13	13:23:09	SELECT book_id, title, embedding, SQRT(...	3 row(s) returned	0.000 sec / 0.000 sec
14	13:23:38	SELECT book_id, title, embedding FROM Bo...	4 row(s) returned	0.000 sec / 0.000 sec

Object Info

Session

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

MySQL Workbench

Local instance MySQL80

FileEditViewQueryDatabaseServerToolsScriptingHelp

Navigator

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Information

No object selected

SQL File 1*

Limit to 1000 rows

61"distance": 0.0

62},

63{

64"book_id": 2,

65"title": "Space Adventures",

66"distance": 0.0866

67},

68{

69"book_id": 5,

70"title": "Astronomy Basics",

71"distance": 0.0346

72}

73]

74{

75"error": {

76"code": "BAD_REQUEST",

77"message": "limit must be a positive integer"

78}

79}

80}

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context HelpSnippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
11	13:22:24	UPDATE Books SET embedding = CASE book_id ...	4 row(s) affected Rows matched: 4 Changed: 4 Wa...	0.031 sec
12	13:22:35	SELECT book_id, title, embedding FROM Books LIM...	4 row(s) returned	0.000 sec / 0.000 sec
13	13:23:09	SELECT book_id, title, embedding, SQRT(...	3 row(s) returned	0.000 sec / 0.000 sec
14	13:23:38	SELECT book_id, title, embedding FROM Bo...	4 row(s) returned	0.000 sec / 0.000 sec
15	13:24:17	{ "query": "stories about space", "results": [{ ...	Error Code: 1064. You have an error in your SQL syn...	0.000 sec
16	13:24:42	{ "query": "stories about space", "results": [{ ...	Error Code: 1064. You have an error in your SQL syn...	0.000 sec

Object InfoSession

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator:..... SQL File 1* x SQLAdditions

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Administration Schemas

Information

No object selected

```
76      "error": {
77        "code": "BAD_REQUEST",
78        "message": "limit must be a positive integer"
79      }
80    }
81  {
82    "query": "space",
83    "results": [
84      {
85        "book_id": 1,
86        "title": "Journey to Mars",
87        "distance": 0.0
88      },
89      {
90        "book_id": 2,
91        "title": "Space Adventures",
92        "distance": 0.0866
93      }
94    ]
95  }
```

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 12	13:22:35	SELECT book_id, title, embedding FROM Books LIM...	4 row(s) returned	0.000 sec / 0.000 sec
✓ 13	13:23:09	SELECT book_id, title, embedding, SQRT(...	3 row(s) returned	0.000 sec / 0.000 sec
✓ 14	13:23:38	SELECT book_id, title, embedding FROM Bo...	4 row(s) returned	0.000 sec / 0.000 sec
✗ 15	13:24:17	{ "query": "stories about space", "results": [{ ...	Error Code: 1064. You have an error in your SQL syn...	0.000 sec
✗ 16	13:24:42	{ "query": "stories about space", "results": [{ ...	Error Code: 1064. You have an error in your SQL syn...	0.000 sec
✗ 17	13:25:14	{ "query": "stories about space", "results": [{ ...	Error Code: 1064. You have an error in your SQL syn...	0.000 sec

Object Info Session